

Effective Context Utilization in Large Language Models: RoPE-Aware Attribution with Cosine Semantic Scoring

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Abstract

Modern large language models advertise large context windows, yet retrieval accuracy degrades for information near the center of long inputs, known as the “lost-in-the-middle” phenomenon [Liu et al., 2024]. Measuring this requires Rotary Position Embedding (RoPE) indices preservation, meaning naive baselines such as token deletion corrupt positional encodings and produce unreliable signals. We evaluate three RoPE-safe methods—substitution ablation, InputX-Grad, and 4-D causal block masking, on a needle-in-a-haystack benchmark (512–2048 tokens, five needle depths, 30 prompts) using Qwen2.5-1.5B-Instruct. We introduce three metrics (EUCR, PWUP, GUD), a cosine-semantic importance variant that captures partial output degradation below the exact-match threshold, and a doc-only ablation mode that removes question-chunk bias from gradient-based ranking. Results show that evidence localization becomes harder as context length increases for all methods: mean needle chunk rank rises from 3.4 to 8.1 for substitution, 3.7 to 11.1 for InputXGrad, and 3.0 to 9.6 for Mask4D between 512 and 2048 tokens. Across methods, PWUP and positive GUD scores indicate consistent recency bias, while doc-only correction reduces question-chunk artifacts, especially for InputX-Grad.

1 Introduction

The ability of transformer-based LLMs to reason over long documents is one of the most commercially consequential properties of modern AI systems. Leading models now support context windows exceeding one million tokens: Gemini 1.5 Pro accepts up to 2M tokens [Gemini Team, 2024], GPT-4.1 supports 1M tokens [OpenAI, 2025], and open-weight models such as Qwen2.5 and LLaMA 3.1 support 128k tokens or

more. These models are deployed for multi-document question answering, legal contract review, medical record summarization, and retrieval-augmented generation (RAG). In all of these settings, users implicitly trust that the model will attend to information wherever it appears in the provided context. That trust is not warranted. Liu et al. [2024] demonstrated this empirically: on multi-document QA, GPT-3.5-Turbo’s accuracy drops by more than 20 percentage points when the relevant document moves from the beginning of the context to the middle. At its worst, performance falls *below closed-book accuracy*—the model does better with *no context at all* than when the answer is buried in the middle of its input. This U-shaped performance curve has since been confirmed across model families and architectures [He et al., 2024], and recent benchmarking shows it persists on the most capable models available today: Bertsch et al. [2025] find that GPT-5, Claude Sonnet 4, and Gemini 2.5 Pro all achieve below 50% accuracy on long-context reasoning tasks at 128k tokens, despite nominally supporting far larger windows. The gap between advertised context capacity and actual context utilization is therefore not a legacy problem. Closing it requires first being able to measure it precisely.

In this work we develop three RoPE-safe attribution methods, a cosine-semantic scoring variant that captures partial output degradation, and a doc-only ablation mode that corrects for question-chunk bias. We evaluate all three on a controlled needle-in-a-haystack benchmark and propose metrics that convert per-chunk importance scores into interpretable diagnostics for positional bias.

Our contributions are:

- We implement three RoPE-safe attribution methods: substitution ablation, InputXGrad, and 4-D causal block masking.
- We propose three metrics: EUCR, PWUP, and

GUD, capturing context breadth, positional distribution, and recency/primacy bias respectively.

- We introduce a **cosine-semantic importance variant** for perturbation methods, using sentence embeddings to score output degradation beyond exact-match sensitivity.
- We introduce a **doc-only ablation mode** that removes the question-chunk bias from gradient-based needle ranking.

2 Related Work

Long-context utilization. Liu et al. [2024] provide a systematic study of positional bias in LLM context usage, testing GPT-3.5, GPT-4, and open-source models on multi-document QA and key-value retrieval. Their finding is a U-shaped accuracy curve as the relevant document position varies, with peak performance at the boundaries and a bottom in the middle. A limitation of their evaluation is that it measures *output* accuracy, meaning that a model could accidentally retrieve the answer from memorized knowledge and receive the same score as one that correctly localizes the relevant passage.

He et al. [2024] attribute the lost-in-the-middle effect to a U-shaped attention bias and propose a calibration mechanism achieving up to 15 percentage-point improvements on NaturalQuestions. Their work focuses on *correcting* the bias; ours focuses on *measuring* and localizing it via chunk-level attribution profiles.

Context attribution. One prior work is Cohen-Wang et al. [2024], who introduce *context attribution*: pinpointing which parts of the context led a model to generate a particular statement. ContextCite learns a linear surrogate model over hundreds of random context ablations, where ablation means *deleting* the source tokens from the sequence, to assign a continuous importance score to each context segment. Their ablation is invalid in RoPE-based models because removing tokens re-indexes every subsequent position, corrupting the rotational angles applied to all downstream query and key vectors and conflating the information effect with a positional artifact. We replace this deletion with in-place substitution and 4-D causal block masking both of which preserve RoPE indices. Second, ContextCite requires hundreds of forward passes per prompt to fit the surrogate model; our methods require $O(N_{\text{chunks}})$ generate calls, and our gradient method requires only a single forward-backward pass.

Another work Wu et al. [2025] takes an interpretability approach, identifying a sparse set of *retrieval heads* responsible for long-context factuality across model families. Their methodology masks out *entire* attention heads; our work operates at the *context chunk* level, asking which input regions matter rather than which model components drive retrieval. The two are complementary.

Perturbation-based attribution and RoPE invalidity. Perturbation methods attribute model outputs by measuring how removing input regions changes the response. In RoPE-based models, token deletion is invalid. One-dimensional padding masks suffer the same flaw: Qwen2-family models apply RoPE before the mask, so masked positions still receive their original positional encodings. Recent work has noted sensitivity to positional index shifts in RoPE models [Picov and Goru, 2026], but to our knowledge no prior work has proposed perturbation-based attribution methods that explicitly preserve all position indices.

3 Background

3.1 Rotary Position Embedding (RoPE) and Attribution Validity

We use standard scaled dot-product attention [Vaswani et al., 2017]. Su et al. [2021] encode position m by rotating query and key vectors in $d/2$ two-dimensional subspaces:

$$R_m^{(j)} = \begin{pmatrix} \cos(m\theta_j) & -\sin(m\theta_j) \\ \sin(m\theta_j) & \cos(m\theta_j) \end{pmatrix}, \quad \theta_j = b^{-2j/d}, \quad b = 10,000. \quad (1)$$

Because $q_m^\top k_n$ depends only on the relative offset $m - n$, position indices are baked into every attention score. **Token deletion is therefore invalid:** removing span $[s, e)$ re-indexes every subsequent token from p to $p - (e - s)$, corrupting the rotational angles for all downstream vectors and conflating information removal with positional distortion. **1-D padding masks are also invalid:** Qwen2 applies RoPE before the mask, and a 1-D mask expands to $[1, 1, 1, n]$, blocking key columns uniformly across all layers and violating the causal structure.

3.2 Attribution Methods

All three methods produce a vector of chunk-level importance scores $\mathbf{s} \in \mathbb{R}^{N_c}$, where $N_c = \lceil L/B \rceil$, L is the sequence length, and $B = 128$ is the chunk size.

Method A: Substitution Ablation. All token IDs in chunk $c_i = [iB, \min((i+1)B, L))$ are replaced with the period token, preserving sequence length and all RoPE indices. Importance is scored against the baseline output:

$$\text{imp}_A(c_i) = \begin{cases} 1.0 & \text{baseline correct, ablated incorrect,} \\ 0.5 & \text{output changes,} \\ 0.0 & \text{otherwise} \end{cases} \quad (2)$$

Method B: InputXGrad. A single forward-backward pass is performed on the concatenation of prompt and greedy-decoded answer. Token-level importance is:

$$\text{imp}_B(t) = \|\nabla_{\mathbf{e}_t} \mathcal{L} \odot \mathbf{e}_t\|_2, \quad (3)$$

where $\mathbf{e}_t$ is the embedding of token t , $\mathcal{L}$ is the cross-entropy loss on the model’s own output, and $\odot$ is element-wise multiplication. Chunk importances are the mean of token importances within each window.

Method C: 4-D Causal Block Masking. For chunk c_i , a four-dimensional additive mask blocks all attention to columns $[s_i, s_i + B)$ while preserving causal structure:

$$M_{q,k}^{(i)} = \begin{cases} -\infty & \text{if } s_i \leq k < s_i + B, \\ -\infty & \text{if } q < k \quad (\text{causal}), \\ 0 & \text{otherwise.} \end{cases} \quad (4)$$

Because the mask is added to the pre-softmax logit matrix *after* RoPE rotations are applied to Q and K , all positional encodings remain intact. Importance is scored identically to Eq. (2).

3.3 Cosine Semantic Importance Variant

Binary correctness scoring (Eq. (2)) fails to detect output degradation that falls short of a wrong answer: the model may produce a shifted response when a causally relevant chunk is ablated, without triggering an exact match failure. To address this we use a cosine-semantic importance score.

Let o_{base} denote the baseline output text and o_i the output after ablating chunk c_i . Using all-MiniLM-L6-v2 [Reimers and Gurevych, 2019] to embed outputs (running on CPU to preserve GPU VRAM), we compute:

$$\text{imp}_{\text{cos}}(c_i) = 1 - \frac{\phi(o_{\text{base}}) \cdot \phi(o_i)}{\|\phi(o_{\text{base}})\| \|\phi(o_i)\|}, \quad (5)$$

where $\phi(\cdot)$ denotes the sentence embedding. Higher values indicate greater semantic drift from the baseline, i.e. higher chunk importance. This variant is applicable to Methods A and C, which produce a per-chunk output text; it is not applicable to Method B, which produces a single output.

3.4 Doc-Only Ablation Mode

In the perturbation methods and InputXGrad, the question chunk receives elevated importance: for Methods A and C, ablating the question makes answering impossible, and for Method B because the gradient of the answer loss flows most strongly through the last attended tokens before generation.

The *doc-only* mode addresses this by assigning zero importance to the question chunk before ranking, restricting the competition to document-region chunks. Formally, if q is the question chunk index:

$$\text{imp}_{\text{doc}}(c_i) = \begin{cases} 0 & \text{if } i = q, \\ \text{imp}(c_i) & \text{otherwise.} \end{cases} \quad (6)$$

This produces *needle rank doc* (NRD), the needle’s rank within the document region only. Both discrete and cosine variants are computed in doc-only mode.

3.5 Evaluation Metrics

Needle Chunk Rank (NCR). The chunk c^* containing the needle tokens is located by token-ID matching. NCR is its rank in the importance vector sorted descending (NCR = 1 is optimal). We report four variants: all-chunk (NCR), all-chunk excluding question (NCR_{-q}), doc-only (NRD), and doc-only cosine (NRD_{cos}).

Effective Utilization of Context Ratio (EUCR). Let $\tilde{s}_i = s_i / \sum_j s_j$ denote normalized importance.

$$\text{EUCR}@ \lambda = \frac{|\{i : \tilde{s}_i > \lambda\}|}{N_c}. \quad (7)$$

EUCR measures context *breadth*: near $1/N_c$ indicates concentration on one chunk; near 1 indicates uniform utilization. We report at $\lambda \in \{0.01, 0.05, 0.10\}$.

Position-Weighted Utilization Profile (PWUP). The context is split into beginning (B), middle (M), and end (E) thirds.

$$\text{PWUP}_r = \frac{1}{|R|} \sum_{i \in R} \tilde{s}_i, \quad r \in \{B, M, E\}. \quad (8)$$

$PWUP_E \gg PWUP_M$ indicates recency bias; $PWUP_B \gg PWUP_M$ indicates primacy bias. The lost-in-the-middle effect predicts $PWUP_M < PWUP_B$ and $PWUP_M < PWUP_E$ simultaneously.

Gradient Utilization Decay (GUD).

$$GUD = \rho\left(\{i\}_{i=0}^{N_c-1}, \{s_i\}_{i=0}^{N_c-1}\right), \quad (9)$$

where ρ is Spearman rank correlation between chunk position index and importance. **Positive GUD indicates recency bias** (later chunks score higher); negative GUD indicates primacy bias. A value near zero indicates no monotonic positional trend.

4 Methodology

4.1 Model and Configuration

All experiments use Qwen2.5-1.5B-Instruct [Yang et al., 2024], a 1.5-billion-parameter instruction-tuned decoder with a native 32k-token context window and RoPE positional encoding with base $b = 10,000$. The model is loaded in `bfloat16` to avoid RMSNorm overflow present in `float16` at this model scale, and with `attn_implementation='eager'` to enable per-layer attention weight extraction and custom 4-D additive masks. Additionally, all experiments were run on a single NVIDIA H200 GPU.

4.2 Prompt Construction

Each prompt is wrapped in the Qwen2.5-Instruct chat template before tokenisation. The haystack is a repeating scientific-text paragraph; two needle types are used uniformly at random:

- **UUID:** “The secret activation code is: X7QK3P1A.”
- **Key-value:** “system_port = 4821”

Both needle types use runtime-generated random values that the model cannot have memorized during pretraining, ensuring any correct answer must come from the context. The haystack is trimmed to the exact target token budget by tokenising, truncating, and decoding; the needle is inserted at word index $\lfloor \delta \cdot |\mathcal{W}| \rfloor$ for depth fraction δ , where $\mathcal{W}$ is the set of whitespace-delimited words after trimming. Cosine importance scores are computed post-hoc using all-MiniLM-L6-v2 [Reimers and Gurevych, 2019] (384-dimensional embeddings, loaded on CPU) after all generation calls are complete.

4.3 Experiment Design

Parameter	Values
Context lengths	512–2048 tokens
Needle depths δ	0.1, 0.25, 0.5, 0.75, 0.9
Examples per config	2
Total prompts	$3 \times 5 \times 2 = 30$
Chunk size B	128 tokens
Max new tokens	64 (greedy decoding)
EUCR threshold λ	0.01, 0.05, 0.10

5 Results and Discussion

We report results over 30 prompts (3 context lengths $\times$ 5 depths $\times$ 2 seeds). All values are averages over the 10 prompts at each context length.

5.1 Needle Chunk Localization (NCR)

Table 1 reports mean NCR in all four variants for each method and context length.

NCR rises sharply with context length for all methods, confirming the lost-in-the-middle hypothesis. At 512 tokens, all methods rank the needle chunk near 3rd on average (out of ~ 4 chunks); by 2048 tokens this rises to 8–11 (out of ~ 16 chunks), approaching a near-random rank.

Figure 1 shows NCR as a function of needle insertion depth. At 512 tokens all three methods show low, stable NCR across depths, with substitution and mask4d reaching near-rank-1 at depth 0.5. At 2048 tokens NCR is uniformly high and the depth-rank relationship becomes irregular: substitution peaks at depth 0.75 (NCR ≈ 12) and InputX-Grad peaks at depth 0.1 (NCR ≈ 16), inconsistent with a simple U-shaped positional bias. The clearest U-shaped pattern—middle depths easiest, boundary depths hardest—appears for InputXGrad at 1024 tokens, where rank dips near depth 0.5 before rising at both extremes. For substitution and mask4d the depth-rank curve is noisier, consistent with binary scoring producing sparse importance vectors where most ablations score 0 regardless of needle position.

Question-chunk correction. The NCR_{-q} column consistently improves NCR by 0.0–1.0 rank positions, most visibly for InputXGrad (3.7 $\rightarrow$ 2.7 at 512 tokens; 11.1 $\rightarrow$ 10.1 at 2048 tokens), confirming that the question chunk systematically occupies the top gradient rank. For Methods A and C the improvement is smaller, as the question chunk is causally necessary regardless of needle position.

Table 1: Mean Needle Chunk Rank by method and context length. Lower = better; ideal = 1. NCR: all chunks. NCR_{-q}: all chunks, question zeroed. NRD: doc region only (from doc-only run). NRD_{cos}: doc region, cosine scoring. InputXGrad cosine variants are N/A (single output, no per-chunk ablated text).

Method	Tok.	NCR	NCR _{-q}	NRD	NRD _{cos}
B – IxG	512	3.7	2.7	2.7	–
	1024	6.1	5.1	5.1	–
	2048	11.1	10.1	10.1	–
C – M4D	512	3.0	2.8	2.8	2.6
	1024	5.9	5.5	5.5	5.1
	2048	9.6	8.3	8.3	7.7
A – Sub.	512	3.4	2.9	2.9	2.6
	1024	5.5	5.0	5.0	5.1
	2048	8.1	8.1	8.1	8.1

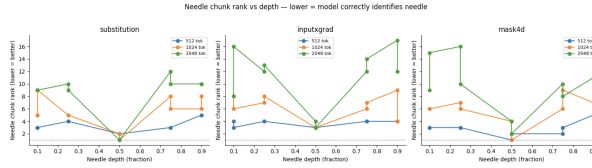


Figure 1: Mean NCR versus needle insertion depth for each method and context length (lower = better; dashed line = ideal rank 1). At 512 tokens, NCR is low and stable across depths. At 2048 tokens, NCR is uniformly high and irregular: substitution peaks at depth 0.75, InputXGrad peaks at depth 0.1, and no simple U-shaped pattern holds across all methods.

Cosine scoring. The NRD_{cos} column consistently lowers rank relative to NRD for Methods A and C (e.g. 2.9 → 2.6 for Method A at 512 tokens; 8.3 → 7.7 for Method C at 2048 tokens), confirming that cosine scoring captures partial output degradation that the exact-match criterion misses.

InputXGrad degradation. InputXGrad degrades fastest at long contexts (NCR = 11.1 vs. 8.1/9.6 at 2048 tokens). As L grows, the gradient signal diffuses broadly, weakening the localized spike at the needle position.

5.2 Doc-Only Ablation Comparison

Figure 2 confirms that doc-only correction consistently reduces NCR across all methods and context lengths, but the magnitude varies systematically. The benefit is largest when the question chunk represents a large fraction of the total chunk count (short contexts) and when the method is gradient-based (InputXGrad), where the question chunk’s position in

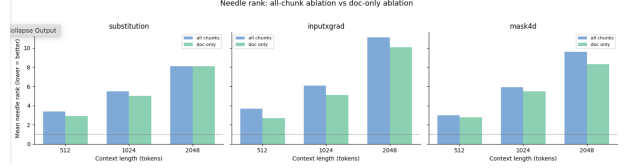


Figure 2: Mean NCR for all-chunk (blue) versus doc-only (green) ablation per method and context length. The gap is largest for InputXGrad at 512 tokens (3.7 → 2.7), where the question-chunk gradient bias is strongest relative to the small total chunk count. For substitution at 2048 tokens the gap is zero (8.1 → 8.1), as the binary signal is too sparse for question-chunk removal to change the ranking.

Table 2: Mean PWUP (Beginning / Middle / End thirds of context) and GUD per method and context length. Positive GUD = recency bias (later chunks scored higher); all values in this table follow the paper’s convention where positive = recency.

Method	Tok.	PWUP _B	PWUP _M	PWUP _E	GUD
B	512	0.213	0.103	0.684	+0.23
	1024	0.242	0.067	0.691	+0.10
	2048	0.219	0.105	0.676	+0.03
C	512	0.168	0.255	0.576	+0.41
	1024	0.238	0.311	0.451	+0.31
	2048	0.289	0.272	0.439	+0.13
A	512	0.277	0.183	0.540	+0.27
	1024	0.434	0.112	0.454	+0.03
	2048	0.378	0.100	0.522	+0.05

the direct gradient path creates the strongest bias. For substitution at 2048 tokens the improvement is zero: most chunks already score 0 under binary scoring, so removing the question chunk does not change the relative ranking of the remaining non-zero entries.

5.3 Positional Bias: PWUP and GUD

Table 2 and Figure 3 report the positional bias profiles at each context length.

Every method and context length exhibits $PWUP_E \gg PWUP_M$, confirming strong recency bias across the board. For Methods A and B the Beginning third also exceeds the Middle, producing the expected U-shaped profile. Method C is a notable exception: $PWUP_M > PWUP_B$ at both 512 tokens (0.255 > 0.168) and 1024 tokens (0.311 > 0.238), suggesting that 4-D masking captures genuine middle-region causal influence that substitution does not. This is consistent with the mask propagating information-blocking effects across all attention heads

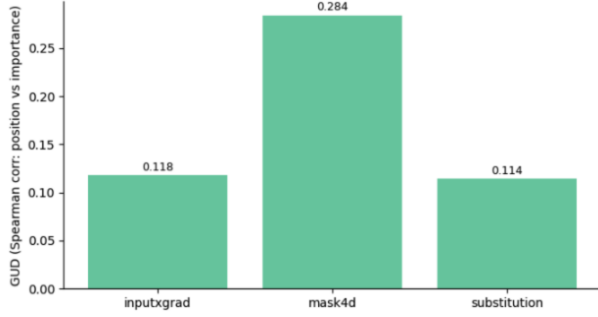


Figure 3: GUD (mean Spearman ρ between chunk position and importance, averaged across context lengths) per method. All three values are positive, indicating recency bias under the paper’s sign convention. Method C shows the strongest recency tendency ($\rho = 0.28$); Methods A and B are weaker ($\rho \approx 0.11$ – 0.12).

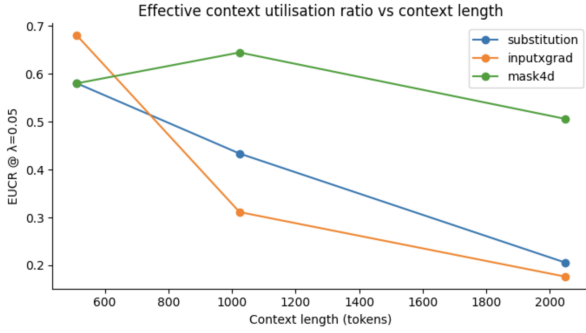


Figure 4: EUCR@ $\lambda=0.05$ as context length increases. Substitution (blue) and InputXGrad (orange) decline monotonically. Mask4D (green) *increases* from 512 to 1024 tokens ($0.58 \rightarrow 0.65$) before declining to 0.51 at 2048 tokens, reflecting that cosine importance under 4-D masking is more distributed at intermediate context lengths.

simultaneously, making middle-chunk ablations more impactful than the embedding-level substitution.

GUD is positive for all three methods (Figure 3), confirming recency bias. Method C has the highest mean GUD ($\rho = 0.28$), which decays toward zero at longer contexts as the binary importance signal spreads across more chunks and the positional trend flattens. Methods A and B show much weaker GUD ($\rho \approx 0.11$ – 0.12): for substitution this reflects the sparsity of binary scores, and for InputXGrad it reflects gradient diffusion distributing importance broadly rather than concentrating it near the end.

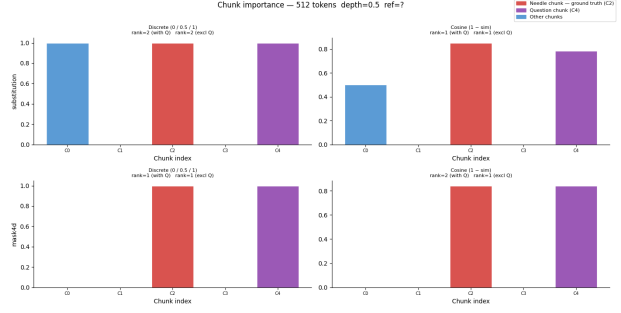


Figure 5: Chunk importance at 512 tokens, depth 0.5 (needle = C2, question = C4). Discrete substitution places a competing chunk (C0) at rank 1; cosine scoring correctly ranks the needle (C2) first for both methods.

5.4 Context Breadth: EUCR

At 512 tokens EUCR@0.05 is 0.68 (InputXGrad), 0.58 (Substitution), and 0.58 (Mask4D). By 2048 tokens these fall to 0.18, 0.21, and 0.51 respectively. Substitution and InputXGrad decline monotonically: for substitution this reflects genuine context sparsity (only one chunk is causally necessary for factual retrieval); for InputXGrad it reflects gradient concentration at the boundaries crowding out the interior. Mask4D behaves differently, with EUCR *rising* from 512 to 1024 tokens before declining. At 1024 tokens the 4-D mask’s attention-blocking affects a broader spread of chunks than at 512, likely because more chunks participate in indirect attention pathways to the answer at intermediate sequence lengths. At 2048 tokens the cosine signal converges toward the boundary-dominated pattern common to all methods, but Mask4D retains the highest EUCR (0.51) by a wide margin, consistent with it spreading causal influence more broadly than in-place substitution.

5.5 Per-Chunk Influence Profiles

Figures 5–7 show discrete and cosine chunk importance profiles for depth-0.5 prompts at each context length, for Methods A and C. The needle chunk is highlighted in red and the question chunk in purple.

The profiles confirm that cosine scoring produces sharper, more discriminative importance peaks than discrete scoring at all context lengths. The advantage is most pronounced at 1024 tokens for Mask4D, where binary scoring is uninformative (flat distribution) but cosine cleanly isolates the needle. At 2048 tokens both methods struggle: discrete Mask4D is completely flat and cosine still ranks the needle below several other chunks, reflecting the fundamental

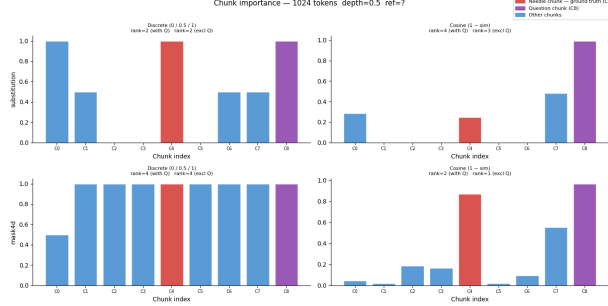


Figure 6: Chunk importance at 1024 tokens, depth 0.5 (needle = C4, question = C8). Discrete Mask4D assigns equal importance to nearly all chunks (rank 4 for needle); cosine Mask4D concentrates on the needle (rank 1 excl. Q), demonstrating that cosine scoring recovers localization where binary scoring degrades.

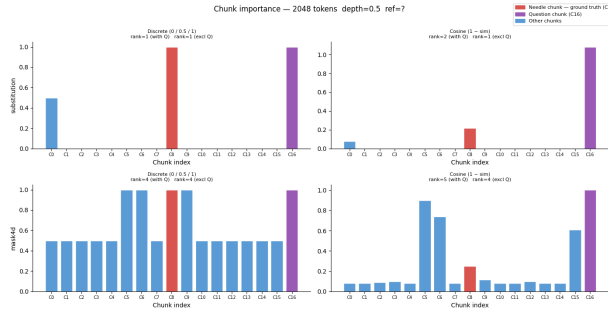


Figure 7: Chunk importance at 2048 tokens, depth 0.5 (needle = C8, question = C16). Discrete Mask4D is nearly flat (all chunks score 0.5); cosine Mask4D partially recovers signal (needle rank 4 excl. Q). Substitution correctly ranks the needle 1st under discrete scoring but loses it under cosine, where the question chunk’s large semantic drift dominates.

difficulty of localizing a single fact in a 16-chunk context with only two seeds per configuration.

5.6 Engineering Decisions and Lessons

bfloat16 vs. float16. Loading Qwen2.5 in float16 produced silent NaN hidden states in RMSNorm layers, causing InputXGrad to produce all-zero gradients. The model outputs appear normal, making this failure invisible without inspecting the GUD column. bfloat16 resolves this completely.

Chat template requirement. Running prompts without the `<|im_start|><|im_end|>` wrapper reduced baseline accuracy substantially; the model was fine-tuned exclusively on templated data.

Explicit EOS token ids. Without specifying both stop token ids (151643 and 151645), the model runs to `max_new_tokens` and decoding strips the stop token, leaving garbage trailing tokens that break substring-match evaluation.

Prefill-then-decode for Method C. Standard `model.generate()` rebuilds the attention mask at every decode step, overwriting the 4-D block mask. A manual prefill-then-token-level-decode loop is required to preserve the mask throughout generation.

5.7 Limitations and Future Work

- **Single small model.** All results are for Qwen2.5-1.5B. Larger models (7B, 72B) and architectures with different positional encoding schemes may exhibit different bias magnitudes.
- **Synthetic haystack.** The repeating scientific paragraph is unrealistically uniform. Real documents have semantic structure that may redistribute importance. Evaluation on SCROLLS or LongBench [Bai et al., 2023] would test generalization.
- **Attention aggregation.** The method is fully implemented but was excluded from the main experiment run. Its inclusion would complete the four-method comparison and allow pairwise agreement analysis between attention-based and gradient/ablation methods.
- **Integrated gradients.** Replacing InputXGrad with full IG [Sundararajan et al., 2017] would provide axiomatic completeness guarantees at the cost of K additional passes.
- **Multi-needle NIAH.** The RULER benchmark [Hsieh et al., 2024] tests retrieval of multiple facts; NCR would need to be replaced with a set-rank metric.

6 Conclusion

We presented three RoPE-safe attribution methods for measuring context utilization in Qwen2.5-1.5B-Instruct, evaluated on a 30-prompt NIAH benchmark across context lengths 512–2048 tokens and five needle depths. Our central technical contributions are: (1) identification and correction of RoPE-invalidity in token deletion and 1-D mask ablation; (2) a cosine-semantic importance variant that captures partial output degradation below the exact-match threshold; and (3) a doc-only ablation mode that removes the systematic question-chunk bias from needle rank computation.

Our results confirm strong recency bias across all methods ($PWUP_E \gg PWUP_M$ at all context lengths) and a consistent NCR degradation from ~ 3 at 512 tokens to ~ 9 at 2048 tokens. The cosine scoring variant often improves needle rank for perturbation methods, especially for Mask4D, but does not improve every setting. InputXGrad degrades fastest at long contexts (NCR = 11.1 at 2048 tokens vs. 8.1 for substitution), consistent with gradient diffusion over softmax distributions that concentrate at context boundaries as L grows.

These findings argue for using perturbation-based methods as ground truth when budget permits, for always applying doc-only correction to remove question-chunk artifacts, and for pairing discrete scoring with a semantic similarity variant to capture the full range of output degradation.

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